Module 3: Problem Discovery and Market Research

Detailed supplement: market research and user research from the Week 3 deck, with a worked discovery case study



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MODULE 3 · HOW TO USE THIS SUPPLEMENT

Coverage map: what is new and what is already in the module

Read this supplement next to the main Module 3 handbook. **Extended** means the handbook has the topic and this PDF adds the deck's example. **New** means the handbook did not have it. The Uber case follows the same method as an end-to-end example.

Topic (Week 3 deck)	Handbook section	Status here
What problem discovery is and why it matters	3.1	Extended: four reasons from the deck
Primary and secondary research, combined	3.2	New: definitions and a worked combination
Market sizing: TAM, SAM and SOM	3.9 (partial)	New: TAM is missing from the handbook
Porter's Five Forces	—	New: five forces and their questions
Competitive analysis	3.9	New: scored comparison and the trade-off lesson
Surveys and their limits	3.4	Extended: fill rate falls after about five questions
Passive and active observation	—	New: how to run each
One-to-many and in-depth interviews	3.3	Extended: format and length
Personas: what they are and are not	3.5	Extended: the test for a useful persona
Worked discovery case: Uber pickup	3.7, 3.8	New: the full method, step by step

NOTE ON THE SOURCE

Some slides in the Week 3 deck are pictures with no readable text. Only the text labels are used here, and no figures are added beyond the source.

WEEK 3 · SECTION 3.1

What problem discovery is, and why it matters

Problem discovery is the process of identifying, understanding and validating the real problems your users face, before you build anything. The deck's point: value is in what gets used, not in what gets built.

Reason	Why it matters
Solve the right problem	Stops effort going to a feature nobody needs
Prevent product failure	Catches the wrong problem before the build
Align with real user need	Gives the team one shared picture of the user
Build user empathy	Makes the PM notice pain that data alone misses

After discovering problems, the next step is to understand the market: is the problem big enough to be worth solving at scale?

Primary and secondary research

Market research tells you whether a problem is worth solving at scale. It gives a strategic view of the opportunity and its competitors.

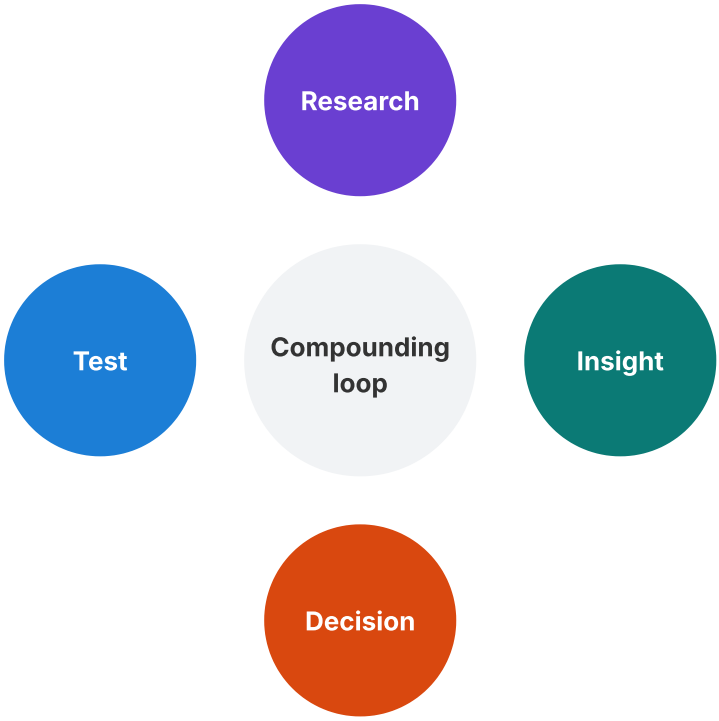
Type	What it is	Why PMs use it
Primary	New data you collect yourself, from users directly	Understand real pain; validate needs
Secondary	Existing data collected by someone else	Benchmark; understand the landscape

Combine both. Example for a fintech app for college students: primary research means interviewing ten students about their money habits; secondary research means reading public trends on digital payments among young adults.

WEEK 3 · DIAGRAM

The discovery loop

Discovery is a loop: each test feeds the next research question. Use the loop to decide what to learn next, not only what to build.



Discovery loop: research, insight, decision, test

NEW · MARKET SIZING

TAM, SAM and SOM

Market sizing estimates the potential of a market in three nested layers. The deck uses a pizza restaurant as the example.



Size the outer ring first.
Then cut to people you
can reach with your model.
Last, the share you can win.

TAM, SAM and SOM as nested circles

Term	Stands for	What it means	Example from the deck
TAM	Total addressable market	Total demand if everyone used the product	All possible customers who could buy pizza in the country
SAM	Serviceable available market	The part you can reach with your model	Customers in your city who would order from you
SOM	Serviceable obtainable market	The share you can realistically win in a few years	The customers one restaurant can serve

WATCH OUT

A big TAM does not prove that your problem is big. Size the people who have the problem, not only the market. Label every assumption and its source.

NEW · MARKET FORCES

Porter's Five Forces

Porter's Five Forces asks how attractive a market is. Each force gives one question to answer.

Force	Question to ask
Threat of new entrants	How easy is it for a new player to enter?
Threat of substitutes	What else can the customer use instead?
Bargaining power of buyers	How much can customers push on price or terms?
Bargaining power of suppliers	How much can suppliers push on cost or terms?
Rivalry among existing competitors	How hard do current players fight for the same customers?

EXTENDED · COMPETITIVE ANALYSIS

Compare competitors on what matters

Competitive analysis means identifying, evaluating and benchmarking competitors: their strengths, weaknesses, positioning and threats. Use it to find a market gap and to prepare your positioning.

The deck's case is an AI transcription startup. Each competitor is scored from 1 (weak) to 4 (strong) on speed, price and error rate, so the trade-offs are explicit. The scores below are illustrative.

Competitor	Speed	Price	Error
Competitor A	1	1	4
Competitor B	4	3	2
Competitor C	3	2	3
Competitor D	2	4	1

The lesson: the table makes your trade-offs visible and shows where the gap is. Pick the dimension where you can be clearly best, then check that the market cares about it.

NEW · RESEARCH METHODS

Passive and active observation

Observation shows how people actually use a product, not how they say they use it.

Method	How to run it	Watch for
Passive observation	Study a user flow chart, pick the flow most users repeat, then watch screen recordings or analytics for the steps where people stop.	Where people hesitate, repeat or give up
Active observation	Ask a user to complete tasks on your product, a competitor's, or a prototype, while you watch and talk with them.	Whether people understand the screen, and can use it alone

- Give goals, not steps: 'how would you book a seat?', not 'click this button'.
- Prepare two or three tasks.
- Tell users it is a prototype if it looks finished, so they do not feel judged.

EXTENDED · SURVEYS

Surveys give a pulse, not a decision

A survey gives a general pulse from users. It helps you decide which direction to explore further, or to drop it. It needs enough responses to show reliable patterns.

- Fill rate falls sharply after about five questions. Keep it short.
- Do not make strategic decisions from survey results alone. Always augment them with a deeper method.

EXTENDED · RESEARCH METHODS

One-to-many and in-depth interviews

Method	Format	Use it to
One-to-many conversation	A moderated group discussion about perceptions and beliefs, across segments; participants talk freely	Hear several views at once and spot shared themes
In-depth interview	One-on-one, 30 to 60 minutes, with a guide you can depart from to follow up	Understand one person's reasons in depth

EXTENDED · PERSONAS

What a persona is, and what it is not

A persona is a model of a user group built from research, not from assumption. The deck's test: a persona that lists only a name, gender, birth year, city and marital status is not useful. A useful persona shows the goal, the context, the pain, and the moment the user decides.

EXAMPLE (FROM THE DECK)

A persona for paying by UPI at fuel stations: who pays, why they pay at a pump, what goes wrong, and what they need to finish quickly.

CASE STUDY · STEP 1

Who we design for

Start by naming the people, not the product. The case splits riders into four groups, each with a different pain:

Segment	Who	Pain	What they need
Tech-park commuters	Daily riders aged 24 to 34 on IT corridors	Many gates; the pin goes to the main entrance; repeated calls when a driver cannot find them	Exit-specific pickup guidance
Mall visitors	Shoppers and diners aged 18 to 35	Many entries and parking zones; crowds at weekends	One easy-to-find spot, shared with the driver
Airport travellers	Frequent flyers aged 22 to 45, time-sensitive	Large, multi-level terminals; zones exist but are not obvious	Clear terminal and gate guidance with little delay
First-time riders	New to ride-hailing or to the city, aged 18 to 30	Unsure of pickup norms; anxious about missing the ride	A guided flow that builds confidence

The shared pattern: an unclear pickup point leads to manual calls or texts, wasted time, and more cancellations. The one core need is a guided, reliable and easy pickup.

CASE STUDY · STEP 2

Sizing the opportunity, with its assumptions

The case uses a top-down size: total market, then the part that is app-based, then the part the team could reach. Every step is written as an assumption you can challenge. The case's headline figure for the wider market is shown in the source; check current industry sources before you quote any market number.

- Write the market in three nested layers: total, app-based, and reachable.
- Label each layer's source and its date.
- Keep one sentence for what would make the number wrong.

WATCH OUT

A large market size does not prove that your specific problem is big. Size the people who have the pickup problem, not only the market.

CASE STUDY · STEP 3

Research goal, method and questions

Goal: understand where, why and for whom the pickup breaks down, and how riders cope today, before designing any fix.

Method	Size	Purpose
Anonymous survey, shared through community groups	25 responses	Find how common the problem is
One-to-one interviews	10 riders	Learn the reasons and the workarounds
Follow-up interviews after a prototype	5 of the 10 riders	Test whether the fix helps

Key questions used:

1. When did you last book a ride, and where?
2. Where do you usually get picked up, and how do you identify the driver?
3. After booking, how often are you unsure where to wait?
4. What is the most frustrating part of the pickup?
5. When a driver cannot find you, what do you do?
6. Have you ever cancelled, or wanted to, because of the pickup? What caused it?
7. Would an app-suggested, easy-to-find spot help?

LESSON

Ask about the last real event, not about what people would like in the future. Note that a survey of 25 people shows a pattern worth testing, not a measured rate.

CASE STUDY · STEP 4

Competitor pickup landscape

For each competitor, ask: what did they ship, what changed for the user, and what gap remains? The source compares several ride-hailing products. The pattern matters more than the names:

Pattern in the market	What it fixes	What it leaves open
Designated pickup zones at airports	Crowd hunting becomes a matched queue	Everyday pickups outside designated zones
Curated pickup points and hotspots	Phone calls become one agreed point	The same gap outside hotspots
Wrong-pin alerts	Reacts after the wrong pin is set	No guidance layer before the mistake
Venue photos and walking guides	Calls become a photo guide	Not yet a spatial route
Landmark-based pins	Raw GPS becomes a landmark	Needs a good landmark list for each place

The gap for this case: a spatial guidance layer that works at everyday places such as malls, tech parks and streets, not only at designated zones.

CASE STUDY · STEP 5

Root cause: find the layer under the symptoms

Symptoms are things you see: multiple entrances, crowds, safety worries, driver idle time. The root cause is the layer the team can change. The case names two:

- **Positioning:** can the app describe a point the rider can find?
- **Coordination:** can rider and driver meet without a phone call?

Multi-entrance buildings and crowds are environmental constraints. They shape the problem, but the team cannot remove them. Solve the layer that the app controls.

FIVE WHYS PRACTICE

Ask why three times: why do riders call? Why is the pin in the wrong place? Why does the app not show a usable point? Write where the chain reaches a cause the team can change.

CASE STUDY · STEP 6

Three opportunities, with evidence and risk

Opportunity	User problem	Evidence (from the survey and interviews)	Expected benefit	Key risk or dependency
1. Smart pickup spot (safe zones and a 3D map)	Wrong spot: positioning	Most riders want a suggested spot; repeated complaints about several exits	Fewer first calls; more first-attempt pickups	Needs venue mapping with partners such as tech parks, malls and airports: real operational effort
2. Compass and proximity guidance	Driver cannot converge on the rider	Nearly every rider calls the driver when the driver cannot find them	Hands-free guidance in the last 100 metres	Needs reliable live direction data; weaker in dense high-rise areas
3. Verified vehicle ID (photo, plate and a light signal)	Doubt about the right car	Riders report uncertainty about the vehicle	Safer and quicker identification	Needs driver adoption and privacy checks

Rank the three by evidence strength, expected benefit and risk. The source puts the first opportunity at the top because it addresses the root layer directly.

CASE STUDY · STEP 7

Metrics to judge the fix

Metric	Type	Why it matters
First-attempt pickup success	Primary	Shows the spot and guidance work the first time
Calls between rider and driver during pickup	Leading	Falls early if coordination improves
Cancellations after driver assignment	Lagging, guardrail	Must not rise because of the change
Driver waiting time	Guardrail	Must not rise because riders are harder to find

Each opportunity needs a test before a full build. A Wizard-of-Oz test (a manual version behind the screen) can check whether riders use the suggested spot before any mapping is built.

Quiz with answers

1. Name the four segments in the case.
   Answer: tech-park commuters, mall visitors, airport travellers and first-time riders.
2. What is the difference between a symptom and a root cause?
   Answer: a symptom is what you see; a root cause is the layer the team can change.
3. Why ask about the last real event?
   Answer: people describe real events more accurately than future wishes.
4. What is a Wizard-of-Oz test?
   Answer: a manual version of a feature, behind the screen, to test demand before building it.
5. Which metric is a guardrail in this case?
   Answer: cancellations after assignment, or driver waiting time.

MODULE 3 · FINAL

Completion checklist

Completion checklist

Done	Item
☐	Read the coverage map and marked the new steps
☐	Wrote four segments with pain and need for one product
☐	Listed the key questions for a short interview
☐	Built a competitor table with the gap left open
☐	Named the root layer and two symptoms
☐	Answered the quiz without notes